THE ADVANCES IN INTERNATIONAL LAW THROUGH THE LAW OF OUTER SPACE

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Undoubtedly, Space Law is the most advanced legal science. It would be incorrect to say that it is the most sophisticated branch of international law as this would amount to ignoring the origin and perspective of the Law of Outer Space.

The concept of Space Law is related to a humanistic philosophy wich began gaining ground until it reached the feeling of peoples and the consensus of lawyers, even before any international agreement was adopted. Its realm goes beyond inter-statal relationships. It considers the welfare of man as the beginning and the end of all human activity, even though a so-called "humanized" technique may be used and notwithstanding the fact that the improvements in machines—including surprisingly refined robots—are trying to outspace the person as main objective of all space activities. The distance of those areas already reached cannot—within the cosmic dimension—reflect the intention of not considering men as the holders of fundamental and non-transferable human rights.

The fact that Space Law is elaborated within the United Nations confirms the above statement. The cycle followed by men—by human beings—as subjects of Law is as follows: MAN-SOCIETY-STATE-INTERNATIONAL COMMUNITY-MANKIND. The United Nations is a forum where the international community expresses its views. Only such community, as a whole, may indicate the principles and determine the rules to govern this law of a universal scope.

Space is the culmination of the concept MAN-SOCIETY. It is a reflection of the present stage and perhaps the definitive one in the development of man within the community. For this reason Space Law is able to determine advances and progress, which amounts to the perfecting of International Law. However, the reverse is not true.

It is understood that in the present paper the term "advance" is used in the sense of improvement. Growth or development may turn out to be defective. It may only be seen as an "advance" when it implies an improvement.

Progress in the field of Space Law is of no small importance and mostly comes under the category *de lege ferenda.* With regard to International Law, and within the field *de lege lata,* the following may be mentioned:

1. Jus Humanitatis

Jus Humanitatis is the Law of, and for, Mankind. Man is its natural holder within the social complex. It is not International Law—which governs international relations—even if Man were considered as a subject of International Law. Neither is it the international community—the third political dimension of Man—but it is the human race as a whole, that is to say, the fourth political dimension of Man.

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This has led many writers to consider Mankind as a legal subject and even state that such a characteristic arises from Law after the entry into force of the Outer Space Treaty.[1]

The Outer Space Treaty repeatedly refers to "mankind," "peoples," instead of "international community", "States", "nations". In the Preamble it is said: "Inspired in the great prospects opening up before *Mankind*. . .", "Recognizing the common interest of all *Mankind*. . ." (italics of the author). In these cases the word "Mankind" appears strengthened by the term "all" and "interest" by the word "common". Thus the intention of the drafters of the Treaty could not have been more explicit.

The Preamble equally states: "Believing that the exploration and use of Outer Space should be carried out for the *benefit of all peoples* irrespective of the degree of their economic or scientific development. . ." (italics of the author). Therefore, "peoples" is strengthened by "all".

Article I provides that "The exploration and use of outer space, including the Moon and other celestial bodies, . . . shall be the *province of all mankind*" (italics of the author).

Because of its nature as universal Law, the Treaty "shall be open to all States" (article XIV). Article V provides that "States Parties to the Treaty shall regard astronauts as *envoys of mankind* in outer space. . ." (italics of the author).

The reference to "prospects", "common interest", "benefits", "the province" are indicating the existence of a legal subject which is not the State, but rather *mankind* or, to be more precise, *HUMANKIND.*

2. Res Communis Humanitatis

International Law has taken the idea of *res communis* from Roman Law and applied it to the different areas in Earth. From the moment that outer space and celestial bodies are subject to a *jus humanitatis,* it is proper to speak of a *res communis humanitatis.* The latin term "humanitatis" is ambivalent and, as already stated, means of and for. We are therefore referring to things—in the legal sense—belonging to and for Humankind.

All the references made in the last paragraph are fully applicable to this concept. It is, however, necessary to add an important provision contained in the Outer Space Treaty (article II): "Outer Space, including the moon and other celestial bodies, *is not subject to national appropriation by claims of sovereignty, by means of use or occupation, or by any other means*" (italics of the author). Thus, if no national occupation on the part of States is possible, it is something common to all Humankind, considered as a whole.

3. The Common Heritage of Mankind

This expression was used for the first time at the United Nations by the present writer in 1967:

[1]Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space, including the Moon and other Celestial Bodies (hereinafter the "Outer Space Treaty"), Jan. 27, 1967, [1967] 18 U.S.T. 2410, T.I.A.S. 610 (effective Oct. 10, 1967).

> Indeed, the paternity of the 'common heritage' concept is, more often than not, attributed to the Permanent Mission of Malta to the U.N. in a Note Verbale of 17 August 1967 (recorded in Doc. A/6695 of 18 August 1967). Yet it is not quite exact. If one looks at the Archives of Publications in the Library of the Palais des Nations at Geneva, it is easy to realize that it was in the U.N. Committee on Outer Space, and not in the Seabed Committee, that the expression 'common heritage' was first used and explained. In this connection resort has been made to Doc. A/AC.105/C.2/SR 75 (Spanish, English and French texts) corresponding to the inaugural session of that year, 19 June 1967, at 3:15 p.m.[2]

Because it was believed that the concept in question would finally be accepted by the international community, we later proposed—in May 1970—the "Draft Agreement on the principles governing the activities of States in the use of the natural resources of the moon and other celestial bodies."[3] Article 1 of this Draft Agreement provides that "The natural resources of the Moon and other Celestial Bodies shall be *the common heritage of ALL MANKIND*" (italics of the author). This is the first international text where the principle appeared. It was later examined in the Seabed Committee and towards the end of 1970 a U.N.G.A. Resolution was adopted—although not a binding agreement—where reference was made to the concept of common heritage which was born in 1954 during the Vth Congress of the International Astronautical Federation, Innsbruck, and applied to the law of outer space. The International Law of the Sea has so far not agreed upon any international text including the principle. It is worthy of mention that, after 26 years of holding priority both in doctrine and as an actual proposal at the U.N. Outer Space Committee (UNCOPUOS) some 13 years ago, the principle of the common heritage of mankind was embodied in the Agreement Governing the Activities of States on the Moon and other Celestial Bodies.[4]

Article XI of this Agreement establishes that "The Moon and its natural resources are *the common heritage of mankind,* which finds its expression in the provisions of this Agreement and in particular in paragraph 5 of this article" (italics of the author).

The above-mentioned paragraph 5 states that "States Parties to this Agreement hereby undertake to establish an international regime, including appropriate procedures, to govern the exploitation of the natural resources of the moon as such exploitation is about to become feasible. This provision shall be implemented in accordance with article XVIII of this Agreement". Article XVIII provides that: . . . A review conference shall also consider the question of the implementation of the provisions of Article XI, paragraph 5, on the basis of *the principle* referred to in paragraph 1 of that article and taking into account in particular any relevant technological developments" (italics of the author).

It is essential to point out that the Agreement, in the above-mentioned article *in fine,* determines the nature of the common heritage concept as a legal principle. The discussion went on for nine years at the U.N. to determine whether it was merely a

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[2]Williams, "The Role of Equity in the Law of Outer Space," 5 Int'l. Rel. 1 (1975).

[3]U.N. Doc. A/AC.105/C.2/L.71 and Corr. 1 (1970). Also, *inter alia,* U.N. Doc. A/AC.105/85, July 3, 1970, Annex II, at 1.

[4]Agreement Governing the Activities of States on the Moon and Other Celestial Bodies, (hereinafter "Moon Agreement"), U.N. GAOR, 34th Sess., Suppl. No. 20, Doc. A/34/20 (1979). See also U.N.G.A. Res. 34/68 (Dec. 14, 1979).

concept—a thesis supported by those who did not agree on the inclusion thereof in the text—or a legal principle, which was the firm position of the author of the first draft and also of all who shared his opinion.

It is not possible within this brief analysis to go into the question of the extraordinary importance that the solemn and unanimous adoption of this legal principle by the international community on 14 December 1979, at the U.N. General Assembly has meant for the development of Space Law and, consequently, of International Law. Yet, reference ought to be made to the Working Paper submitted by Argentina on 17 April 1973 at the Legal Sub-Committee of U.N. COPUOS[5]. Without any claim to exhaust other inherent points of substance, the following may be stated:

> A realization on the part of all States and peoples that they are entitled to the benefits derived from the principles and norms established for outer space and celestial bodies;
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> The need to link to the exploration and use of outer space the exploitation thereof;
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> The search for profit, with an attempt to share its results;
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> Equitable sharing of the profits derived;
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> Consideration of the needs and interests of developing countries;
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> Supervision of this activity with a view to equitable distribution;
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> The institution of an international regime;
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> The establishment of appropriate procedures for such a regime;
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> The existence of an international machinery or an international authority to give effect to all the expectations that have been voiced.

The major merit of replacing the vague expression "province of all mankind" by the more meaningful expression "common heritage of all mankind" is that in so doing one has specified the commencement of an action, replacing an abstract statement by a means of operating, within a specified legal framework.

With respect to the international regime—twice mentioned in the document in question—reference may be made to my opinion given to my distinguished friend, Mrs. Eilene Galloway in a letter of 31 January 1980 which is quoted with her permission.

> It is rather dangerous to crystallize in a definition the principle involved in a concept which is being just born in a new domain of Space Law, such as "the common heritage of mankind", as it was established in the Moon Agreement. As for the "international regime" concerned, I dare say it is not a matter of definition; I feel it must be the outcome of the implementation of the guidelines set forth in the agreement.[6]

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[5]U.N. Doc. A/AC.105/115, Annex I, at 29-30-31 (1973).

[6]The Council of Advanced International Studies, Ambassador Cocca's correspondence, No. 384 (Jan. 1980).

Finally, in my view, it is preferable to speak of the *common heritage of mankind,* so as to give a more spiritual connotation to the newly adopted legal principle.

4. Representation of Mankind in Outer Space and Celestial Bodies

The importance of the provision in article V of the Outer Space Treaty: "States Parties to the Treaty shall regard astronauts as envoys of mankind. . ." lies in the fact that astronauts have been invested with the legal representation of mankind in outer space and celestial bodies. No other former representation has ever been as wide, and—politically—it goes beyond the most audacious ambition. On the other hand, it should be borne in mind that it has been conferred by the international community as a whole, in a public assembly, by unanimity and acclamation. Also, there is no precedent on any similar means of conferring a mandate.

5. Exploration and Use for the Benefit and Interest of all Peoples

The Preamble to the Outer Space Treaty speaks of "the benefit of all peoples irrespective of the degree of their economic or scientific development". Article I confirms the Preamble: "The exploration and use of outer space, including the moon and other celestial bodies, *shall be carried out for the benefit and in the interests of all countries,* irrespective of their degree of economic or scientific development, *and shall be the province of all mankind*" (italics of the author). It should be recalled that the Declaration of Legal Principles which preceded the Treaty required that all activities of exploration and use shall be carried out for the benefit and in the interest of all mankind.[7]

The use of the verb "shall be" is of great importance in both cases. And all of us who have some time taken part in the elaboration of this Treaty certainly are aware of the difficulties of reaching consensus on the tense used within the text.

6. International Cooperation as a Requisite for the Lawfulness of Activities of Exploration and Use

The term "international cooperation" appears several times throughout the text of the Outer Space Treaty. The tense of the verb clearly indicates that it amounts to something more than an aspiration: it is an obligation undertaken by States Parties to the Treaty. Article I, paragraph 3, with regard to scientific investigation establishes that "States *shall facilitate and encourage* cooperation in such investigation" (italics of the author). Article III provides that States Parties "shall carry on. . . [activities] promoting international cooperation and understanding". Article X states: "In order to promote international cooperation in the exploration and use of outer space. . . the States Parties to the Treaty shall consider. . .". Similarly, article XI provides: "In order to promote international cooperation in the peaceful exploration and use of outer space. . . States Parties agree. . .". Article XII states that, "All stations, installations, equipment and space vehicles on the moon and other celestial bodies shall be open. . .".

[7]U.N.G.A. Res. 1962/XVIII (Dec. 13, 1963).

From the foregoing observations one may fairly conclude that the participants in the CNIE-UNESCO Seminar on "The teaching of International Law as applied to outer space and space communications" (Buenos Aires, August 1972) were certainly right in declaring that international cooperation was a legal obligation arising from the Outer Space Treaty, an obligation which conditioned the lawfulness of space activities.[8]

7. Freedom of Exploration and Use of Outer Space and Celestial Bodies and Banning of National Appropriation

When the exploration and use of outer space is carried out for the benefit and in the interest of Humankind and international cooperation is a necessary assumption for outer space activities—which may never be carried out for the benefit of one State or group of States but only, for the general benefit of mankind—it obviously follows that outer space and celestial bodies are open for exploration and use. Hence the reading of article I: "Outer Space, including the moon and other celestial bodies, shall be free for exploration and use by all States without discrimination of any kind, on a basis of equality and in accordance with international law, and there shall be free access to all areas of celestial bodies" and "There shall be freedom of scientific investigation. . ."

The recognition of these rights to States entails certain obligations: every State is bound to respect the rights and interests of other states, which amounts to speaking of the rights and interests of mankind which is acting through the State.

The principle of non-appropriation, established in article II of the Outer Space Treaty, is a necessary complement to the principle of freedom: "Outer space, including the moon and other celestial bodies, is not subject to national appropriation by claim of sovereignty, by means of use or occupation, or by any other means". This principle was first enunciated in Resolution 1721 (XVI) of the U.N. General Assembly[9] and was later included in the Declaration of Legal Principles contained in Resolution 1962.

8. The Predetermined Legal Framework

Article I, paragraph 2 and article III of the Outer Space Treaty establish the application of international law and the U.N. Charter to all outer space activities. Henceforth and so far, the determination of a precise legal framework has remained the goal of the drafters of these treaties. If we look at the recently adopted Agreement Governing the Activities of States on the Moon and other Celestial Bodies, it may be seen that it includes the traditional mention of International Law and the U.N. Charter, later adding: "and taking into account the Declaration on Principles of International Law concerning Friendly Relations and Cooperation among States in accordance with the Charter of the United Nations" (October 24, 1970). Most significant is a specific reference to the law of outer space contained in the Preamble to the said Agreement where the four preceding Conventions are also referred to.

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[8]La Ensenanza del Derecho Internacional aplicado al espacio y a las communicaciones espaciales, CNIE-UNESCO Seminar, Buenos Aires, August 1972, at 81.

[9]U.N.G.A. Res. 1971/XVI (Dec. 20, 1961).

9. Demilitarization

A complete demilitarization is established for the Moon and other Celestial Bodies, according to the second paragraph of article IV of the Outer Space Treaty. There is a regime of partial demilitarization for objects carrying nuclear weapons or any other kinds of weapons of mass destruction, install such weapons on celestial bodies, or station such weapons in outer space (article IV, first paragraph).

Article III of the Moon Agreement provides that "Any threat or use of force or any other hostile act in the moon is prohibited. It is likewise prohibited to use the moon in order to commit any such act or to engage in any such threat in relation to the Earth, the Moon, spacecraft, the personnel of spacecraft or man-made space objects" (paragraph 2). "States Parties shall not place in orbit around or other trajectory to or around the moon objects carrying nuclear weapons or any kinds of weapons of mass destruction or place or use such weapons on or in the moon" (paragraph 3). "The establishment of military bases, installations and fortifications, the testing of any type of weapons and the conduct of military manoeuvres on the moon shall be forbidden" (paragraph 4).

10. International Responsibility and Full Compensation

International responsibility, that is, the liability of States and international intergovernmental organizations, is an important step in the pursuance of a better legal balance among the rights and obligations normally recognized among States. In the field of Space Law, the State is the first responsible.

The principle of State liability appears in the 1962 U.N.G.A. Declaration (XVIII). It is embodied in article VI of the Outer Space Treaty as follows: "States Parties to the Treaty shall bear international responsibility for national activities in outer space, including the moon and other celestial bodies, whether such activities are carried on by governmental agencies or by non-governmental entities. . ." When activities are carried on by an international intergovernmental organization, the "responsibility for compliance with this Treaty shall be borne both by the international organization and by the States Parties to the Treaty participating in such organization". Article VII states that "Each State Party to the Treaty that launches or procures the launching of an object into outer space. . . and each State Party from whose territory or facility an object is launched, is internationally liable for damage to another State Party to the Treaty or to its natural or juridical persons . . ."

The Moon Agreement repeats the principle in article XIV, paragraph 1. Paragraph 2 provides that "States Parties to the Treaty recognize that detailed arrangements concerning liability for damage caused on the moon, in addition to the provisions of the Outer Space Treaty and the Convention on International Liability for Damage caused by Space Objects, may become necessary as a result of more extensive activities on the moon. Any such arrangements shall be elaborated in accordance with the procedures provided for in article XVIII of this Agreement".

The principle of "full compensation" appears in the Convention on International Liability for Damage caused by Space Objects.[10] The Preamble recognizes in its fourth

[10]Convention on International Liability for Damage caused by Space Objects, March 29, 1972, [1973] 24 U.S.T. 2389, T.I.A.S. 7762 (effective Oct. 9, 1973).

paragraph "the prompt payment under the terms of this Convention of a *full* and equitable measure of compensation to victims of such damage" (italics of the author). The nucleus of this Convention, however, is the non-limitation of the quantum: "in order to provide such reparation in respect of the damage as will restore the person, natural or juridical, State or international organization on whose behalf the claim is presented to the condition which would have existed if the damage had not occurred" (article XII).

11. The prevention of contamination of outer space and the Earth

It is hardly surprising to speak of contamination and pollution, particularly after the 1972 Stockholm Conference where general consensus existed on the serious damage threatening the health of humanity and life on this planet due to contamination caused by man himself.

Space Law has, in this field, important precedents: in 1965 the Scientific and Technical Sub-Committee raised the problem in a recommendation attached to a report submitted by COPUOS to the XX Session of the General Assembly.

Resolution 2130 (XX) adopted the above-mentioned recommendation concerning the struggle against potentially harmful interference of space activities.[11]

The Outer Space Treaty has freed outer space from radioactive contamination in article IX. A possible contamination by Earth substances in outer space has also been foreseen, as well as the prevention of eventual damage to the Earth environment due to the introduction of extra-terrestrial matter.

Article IX provides that ". . . States Parties to the Treaty shall pursue studies of outer space, including the moon and other celestial bodies, and conduct exploration of them so as to avoid their harmful contamination and also adverse changes in the environment of the Earth resulting from the introduction of extra-terrestrial matter and, where necessary, shall adopt appropriate measures for this purpose". This article also establishes the right and obligation of holding consultations at international level when the activity of a State would cause potentially harmful interference with the activities or experiments of others or its own.

These innovations undoubtedly mean an advance in the field of legal sciences. The law of outer space and celestial bodies has proudly gained a place within the realm of juridical knowledge and, as a precursor in many aspects, holds a legitimate paternity with respect to new disciplines, such as the law of the seabed and ocean floor which, in a first generation, borrowed the concept of the common heritage of mankind from the law of outer space. This principle is now being taken on by international environmental law and international energy law.

[11]U.N.G.A. Res. 2130/XX (Dec. 20, 1965).